

Comprehensive Pipeline Analysis Protocol

De Novo, Mosaic and Somatic Mutation Analysis from Trio WGS / WES Sequencing Data

Version 2.0 · Standard Operating Procedure · supersedes v1.0

Field	Value
Scope	Germline DNMs, post-zygotic and parental mosaicism, functional annotation and provisional ACMG/AMP classification, and somatic mutation analysis (tumour or affected tissue vs blood; clonal haematopoiesis screen) for WGS, WES and large panels
Inputs supported	FASTQ, BAM/CRAM, or joint trio VCF; optional tumour / affected-tissue BAMs
Reference build	GRCh38 no-alt analysis set (primary); GRCh37 supported (PAR coordinates handled by build)
Family configuration	Trio (proband + both biological parents); quads and multi-trio pedigrees processed trio by trio
Reference implementation	<code>trio_dnm_pipeline/</code> — Nextflow DSL2 workflow + <code>trio-dnm</code> Python engine (stdlib only, unit-tested)
Expected runtime	~12–24 h per 30× WGS trio on a 16-core node (germline); +8–16 h per tumour–normal pair (somatic)
Validation requirement	All reportable candidates orthogonally validated (Sanger, ddPCR, amplicon-UMI or long-read)

0. Revision Summary (v1.0 → v2.0)

The v1.0 protocol was reviewed line by line against current tool documentation and the DNM literature. Three commands would fail or silently corrupt results as written, one QC step was not computable, and several biology rules missed whole variant classes. All are corrected below; section numbers refer to v2.0.

0.1 Corrections to v1.0

#	v1.0 location	Problem	Severity	v2.0 fix
1	5.1 HaplotypeCaller	<code>--output-mode EMIT_VARIANTS_ONLY</code> combined with <code>-ERC GVCF</code> removes the reference-confidence blocks. Parental hom-ref status — the core evidence for a DNM — is then unknown at every site the parent did not vary.	Critical	Option removed (§5.1)
2	5.1 HaplotypeCaller	Inline <code># PCR-free libraries</code> after <code>--pcr-indel-model NONE</code> with no trailing backslash ends the shell command; the remaining arguments run as a separate (failing) command.	Critical	Comments moved out of continued lines (§5.1)
3	5.2 GenotypeGVCFs	<code>--use-new-qual-calculator</code> no longer exists in GATK 4.5 (new QUAL model is the default since 4.1); the command errors.	Critical	Flag removed (§5.2)

#	v1.0 location	Problem	Severity	v2.0 fix
4	7.6 Layer 6	Kinship cannot be computed from the filtered DNM set: DNMs are by definition absent from parents, and ~70 sites carry no relatedness information.	High	Replaced by parental-leakage, raw Mendelian-error and allele-balance checks (§7.6)
5	7.1 Layer 1	No sex-chromosome logic. A male proband's chrX DNM is hemizygous (GT 1, VAF $\approx$ 1.0) and fails the 0/1 and VAF 0.30–0.70 rules; the father is irrelevant outside the PARs, chrY DNMs depend only on the father.	High	Ploidy-aware rules by PAR/non-PAR region (§7.1)
6	10.1 VEP	<code>--pick_order</code> does nothing without <code>--pick/--flag_pick*</code> ; CADD/REVEL/AlphaMissense/SpliceAI plugins need data-file arguments; gnomAD v4 has no <code>AF_popmax</code> (renamed <code>grpmax</code>).	High	Working command with plugin files and <code>--flag_pick_allele_gene</code> (§10.1)
7	5.3	Intersecting GATK-PASS with DeepVariant-PASS <i>before</i> DNM detection discards true DNMs (especially indels) that one caller filters.	Medium	Union for candidate discovery; agreement used as evidence in the consensus tier (§5.3, §6.5)
8	6.4 DeepTrio	Described as "trained on PacBio HG002 trios". DeepTrio ships separate Illumina WGS, WES and PacBio models trained on GIAB trios.	Medium	Description corrected; model chosen by data type (§6.4)
9	7.1 / 9	Parental AD _{alt} $\leq$ 1 silently discards variants inherited from a mosaic parent — the cases with the highest recurrence risk.	Medium	Dedicated parental-mosaic track (§9.4)
10	9.3	VAF bins mislabelled ("0.05–0.30 germline-mosaic boundary"); at 30 $\times$ a VAF of 0.30–0.45 is not distinguishable from a heterozygote by eye.	Medium	Binomial test against 0.5 plus Wilson CI (§9.3)
11	8.2	<code>PhaseByTransmission</code> is GATK3-only and cannot phase a DNM (it is absent from both parents). Only ~20–40 % of short-read DNMs are phaseable.	Medium	Read-backed phasing with unfazed; report fraction with CI and n (§8.2)
12	10.2 tiers	REVEL $>$ 0.7 and CADD $\geq$ 25 are uncalibrated; LOEUF $<$ 0.35 is a gnomAD v2 decile cut-off; SpliceAI $\geq$ 0.5 misses the ClinGen PP3 range ($\geq$ 0.2).	Medium	ClinGen-calibrated thresholds; gnomAD v4 LOEUF $<$ 0.6 (§10.3–10.8)
13	3.2	"Kinship $\approx$ 0.25" and somalier "relatedness $\approx$ 0.5" describe the same parent–child relationship on different scales; thresholds were ambiguous.	Low	Relatedness and IBS0 stated explicitly (§3.2)
14	5.3	Hard filters omitted GATK's <code>QUAL < 30</code> criterion.	Low	Added for SNVs and indels (§5.3)
15	7.3	"Segmental duplications (UCSC Self-Chain)" conflates two tracks.	Low	GIAB genome stratifications used as the canonical mask set (§7.3)
16	8.2 vs 13	Paternal-fraction alarm threshold inconsistent (65 % vs 60 %).	Low	Single threshold 65–90 % (§8.2, §14)

#	v1.0 location	Problem	Severity	v2.0 fix
17	4.1	Read-group line omitted PU although the text requires it.	Low	Added (§4.1)
18	6.1	INFO tag is <code>hiConfDeNovo</code> , not <code>hi_conf_denovo</code> .	Low	Corrected (§6.1)

0.2 Additions in v2.0

- **Bayesian trio posterior computed in-pipeline** from the joint VCF genotype likelihoods, independent of the external callers (§7.5).
- **Parental-mosaic and post-zygotic tracks** with statistical VAF classification and clonal-haematopoiesis (CH) confounder screening (§9).
- **Expanded functional prediction:** LOFTEE and NMD escape, AlphaMissense, REVEL, CADD v1.7, BayesDel, PrimateAI-3D, ESM1b, EVE, MetaRNN, SpliceAI (Pangolin optional), UTRAnnotator, conservation, ENCODE cCREs, protein domains and gene constraint including `s_het` (§10).
- **Provisional ACMG/AMP classification** with ClinGen-calibrated PP3/BP4 and the Bayesian point system; ranked prioritisation with phenotype scores (§10.7–10.9).
- **Cohort statistics:** gene-level DNM enrichment (DeNovoWEST / denovolyzeR) and signatures (§10.10).
- **New Stage 9 — somatic mutation analysis:** tumour or affected tissue vs blood, trio-aware germline subtraction, copy number, clonality, TMB, signatures, drivers, germline + somatic two-hit detection, and CH screening (§11).
- **Reference implementation** mapping Stages 1–9 to Nextflow processes and tested `trio-dnm` commands; Stage 10 (orthogonal validation) is laboratory work (§15).

0.3 Review of the v2.0 draft

An independent verification pass checked the v2.0 draft against the reference implementation, current tool documentation and the primary literature. It corrected the REVEL BP4 Strong threshold (≤ 0.016 , with BP4 Very strong ≤ 0.003 ; §10.3), the source of per-sample strand evidence (a candidate-site pileup, because the joint VCF carries no FORMAT/SB; §7.2), the consensus tiers when no external caller is run (§6.5), mosaic routing (binomial test **and** $\text{VAF} < 0.40$; §7.1), the depth floor for hemizygous fathers (§7.1), the gnomAD v4.1 field names (`AF_grpmax_joint`) and several workflow commands (DeNovoGear input, SpliceAI window, stub dry-run, section cross-references). It also added the `SNV QUAL < 30` filter, the Layer-2 FILTER check, evaluation of DeepTrio-only candidates, the Mendelian-error stop-gate in `trio-dnm call` and parental mosaics genotyped as heterozygous (§9.4).

1. Purpose and Scope

This protocol defines a standardised workflow for identifying de novo mutations (DNMs) — variants present in a proband but absent from both biological parents — from trio short-read sequencing, and for characterising their functional consequence. Because >99 % of raw Mendelian-violation calls are technical artefacts, the workflow is built around independent lines of evidence and explicit, auditable filter layers.

The pipeline integrates: (i) pre-flight sample QC with hard stop-gates; (ii) joint trio calling with two independent engines; (iii) multi-caller DNM detection plus an in-pipeline Bayesian posterior; (iv) a layered filter cascade; (v) biological sanity checks; (vi) post-zygotic and parental-mosaic tracks; (vii) functional annotation, calibrated pathogenicity evidence and prioritisation; (viii) somatic mutation analysis; and (ix) orthogonal validation.

Out of scope (require dedicated companion pipelines): structural and copy-number DNMs (e.g. GATK-SV, Manta + CNVnator trio mode), mitochondrial heteroplasmy, short-tandem-repeat expansions (ExpansionHunter), and long-read-only analyses. Somatic analyses are now in scope (§11).

2. Inputs, References, and Software Environment

2.1 Required inputs

- Paired-end Illumina FASTQ or analysis-ready BAM/CRAM for proband, father and mother — minimum 30× WGS or 100× mean on-target WES.
- Sample sheet (CSV): `family_id`, `sample_id`, `role` (proband | father | mother | tumor | tissue), `sex`, then either an analysis-ready BAM/CRAM or paired FASTQ with library/lane metadata (FASTQ is processed by Stage 2), and optional `dna_source` (`dna_source=lcl` marks lymphoblastoid cell-line DNA, §3.3). `conf/samplesheet.example.csv` shows the columns.
- Pedigree in PLINK PED format with IDs identical to BAM `@RG SM` tags.
- Parental ages at conception (strongly recommended: required for the age-adjusted DNM expectation, §8.1).
- HPO terms for the proband (for phenotype-driven ranking, §10.9).
- Optional, for §11: tumour or affected-tissue BAM, tumour purity estimate, FFPE status.

2.2 Reference resources

Resource	Source / version	Purpose
Reference FASTA	GRCh38 no-alt analysis set (+ .fai, .dict)	Alignment and calling scaffold
Known sites	dbSNP 156, Mills/1000G gold indels, 1KG phase 3 SNVs	BQSR
Population AF	gnomAD v4.1 joint (exomes + genomes) — <code>AF_joint</code> , <code>AF_grpmax_joint</code>	Population filter; PM2/BS1/BA1
Region stratifications	GIAB genome stratifications v3.x (low mappability, segdups, tandem repeats, homopolymers), ENCODE blacklist v2	Hard masks (§7.3)
Consequence	Ensembl VEP 113 cache (MANE Select v1.4), LOFTEE, NMD and UTRannotator plugins	Transcript consequence, LoF confidence
Missense predictors	AlphaMissense, REVEL, CADD v1.7, dbNSFP 4.x (BayesDel, PrimateAI-3D, ESM1b, EVE, MetaRNN)	PP3/BP4 and ranking
Splicing	SpliceAI masked pre-computed scores (Illumina, ±50 bp, GENCODE v24); local SpliceAI (-D 500) or SpliceAI-lookup for ±500 bp; Pangolin optional	Splice PP3/BP4, Tier 3
Conservation / regulatory	phyloP (100-way vertebrate / 447-way mammal), GERP++, ENCODE SCREEN cCREs v4	Non-coding prioritisation
Gene constraint	gnomAD v4.1 pLI / LOEUF / missense Z, GeneBayes s_het, ClinGen dosage (HI) scores	Gene-level prioritisation, PVS1 applicability
Clinical knowledge	ClinVar (monthly), OMIM, GenCC, Gene2Phenotype (DDG2P), HPO	Disease association, phenotype match

Resource	Source / version	Purpose
Somatic resources	Mutect2 af-only gnomAD, panel of normals, small_exac_common; COSMIC CGC, cancerhotspots, OncoKB (licensed)	§11
Signatures	COSMIC SBS v3.4 (GRCh38; SBS40 split into SBS40a/b/c)	Germline and somatic spectra

2.3 Software environment

Every tool runs in a container (Docker/Singularity) declared in `nextflow.config`; pin each image by digest once validated. Versions below are the baseline; verify availability before pinning. `trio-dnm` tasks stage the `trio_dnm` package into each task directory and put it on `PYTHONPATH`, so they run in any Python ≥ 3.10 image (default `python:3.11`) or in the image built from the provided Dockerfile. Every resource file (with any index next to it) is staged into the tasks as a Nextflow `path` input, so it is visible inside Docker/Singularity containers without bind mounts (see the header of `nextflow.config`).

Stage	Primary tool(s)	Baseline version
FASTQ QC / trimming	FastQC, fastp, MultiQC	0.12.1 / 0.23.4 / 1.25
Alignment	BWA-MEM2 (or DRAGEN)	2.2.1
Sort / dedup / BQSR	samtools; GATK MarkDuplicates (or MarkDuplicatesSpark), BaseRecalibrator	1.21 / 4.6
Germline calling	GATK HaplotypeCaller + GenotypeGVCFs; DeepTrio + GLnexus	4.6 / 1.8 / 1.4.1
DNM callers	GATK CGP + PossibleDeNovo, DeNovoGear, TrioDeNovo, DeepTrio, trio-dnm posterior	4.6 / 1.1.1 / latest / 1.8 / 2.0
Parent of origin	unfazed	1.0.x
Mosaic	Mutect2 (tumour-only mode), MosaicForecast, DeepMosaic (optional)	4.6 / 0.0.1 / 1.x
Somatic	Mutect2, Manta → Strelka2, DeepSomatic, CNVkit (or PURPLE/FACETS)	4.6 / 1.6.0 / 2.9.10 / 1.8 / 0.9.11
Annotation	Ensembl VEP (+ plugins), vcfanno, bcftools	113 / 0.3.5 / 1.21
Sample QC	mosdepth, VerifyBamID2, somalier	0.3.10 / 2.0.1 / 0.2.19
Post-calling engine	trio-dnm (this protocol)	2.0.0
Workflow engine	Nextflow DSL2	≥ 23.10

3. Stage 1 — Pre-flight Sample Quality Control

Sample swaps, non-paternity and contamination silently corrupt every downstream DNM call. Stage 1 must pass before variant calling; `trio-dnm qc-gate` evaluates the gates and exits non-zero on failure, which halts the workflow. The Mendelian-error gate needs the joint call set and is evaluated by `trio-dnm call` (§3.3). All gate thresholds are in the `qc` section of the configuration (`qc-gate --config`, `call --config`).

3.1 Sequencing-level QC

- FastQC on all FASTQ, aggregated by MultiQC; flag abnormal GC, adapter content $> 5\%$, or Q20 base fraction $< 90\%$.

- fastp: adapter auto-detection, poly-G trimming (NovaSeq/NextSeq), minimum length 50 bp, low-complexity filter.
- mosdepth after alignment: WGS mean $\geq 28\times$; WES $\geq 80\times$ on target with $\geq 90\%$ of targets at $\geq 20\times$; coverage CV < 0.30 for WES. `qc-gate` checks the mean depth (WARN below $28\times$ WGS / $80\times$ WES / $200\times$ panel); target uniformity is reviewed in the MultiQC report.

3.2 Sample identity and relatedness

- VerifyBamID2 FREEMIX < 0.02 pass; $0.02\text{--}0.05$ investigate; > 0.05 fail.
- somalier `relate`: parent–child **relatedness 0.40–0.60** (kinship coefficient ≈ 0.25) **and IBS0 ≈ 0** ($< 0.5\%$ of informative sites). Low IBS0 is the discriminating signal — full siblings also show relatedness ≈ 0.5 but have IBS0 > 0 .
- Parents: relatedness < 0.10 unless consanguinity is documented (it changes the prior on inherited homozygosity).
- Sex inferred from chrY depth ratio and chrX heterozygosity must match the pedigree for **all three members** (PED father male, mother female): a father/mother label swap leaves both parent–child relatedness values at ≈ 0.5 and is caught only here. A sex that cannot be inferred is a WARN (possible sex-chromosome aneuploidy such as XXY or XO, or low coverage).
- Raw Mendelian error rate among trio-variant autosomal sites, computed by `trio-dnm call`: typically $0.5\text{--}2\%$; $> 5\%$ indicates a swap or contamination. Over ≥ 1000 informative sites (`qc.mie_min_sites`) this halts `call` (§3.3); below that (small panels) the gate is not applied, but the rate and a warning are still reported in the call summary and HTML report.

3.3 Stop-gates

Gate	Fail condition	Action
Contamination	FREEMIX > 0.05 in any member ($0.02\text{--}0.05$: WARN)	Halt; re-extract or re-sequence
Sex	Inferred $\neq$ reported sex for any trio member (<code>qc-gate</code>); inferred sex unknown = WARN	Halt; resolve swap or PED error. On WARN resolve possible sex-chromosome aneuploidy before interpreting chrX/chrY calls
Parentage	Parent–child relatedness outside $0.40\text{--}0.60$ or IBS0 $> 0.5\%$	Halt; confirm pedigree (consent-appropriate handling of non-paternity)
Depth	Mean depth $< 28\times$ WGS / $80\times$ WES / $200\times$ panel	WARN; top up or interpret sensitivity with caution
Mendelian errors	Raw autosomal rate $> 5\%$ over ≥ 1000 informative sites (<code>trio-dnm call</code> , after joint calling)	Halt: <code>call</code> writes its outputs and exits with code 3; <code>--no-halt</code> overrides (justify in the QA log)
DNA source	Lymphoblastoid cell-line DNA (samplesheet <code>dna_source=lcl</code> or <code>qc-gate --lcl SAMPLE</code>)	WARN; proceed with flag: LCL culture introduces somatic "DNMs" at low VAF

4. Stage 2 — Read Processing, Alignment and BQSR

4.1 Alignment

```
bwa-mem2 mem -t 16 -K 100000000 -Y \
  -R "@RG\tID:${fc}.\tSM:${sm}\tLB:${lb}\tPL:ILLUMINA\tPU:${fc}.\tlane}.\tbarcode}" \
  ${REF} ${R1} ${R2} | samtools sort -@ 8 -m 2G -o ${id}.sorted.bam -
```

-Y soft-clips supplementary alignments; -K fixes the batch size so output is deterministic across thread counts. In the reference workflow `modules/align.nf` runs this stage when the sample sheet gives FASTQ: fastp and FastQC, BWA-MEM2 with the full read group above, duplicate marking and BQSR. BAM/CRAM input skips it.

4.2 Duplicate marking

GATK MarkDuplicates or MarkDuplicatesSpark (optical distance 2500 for patterned flow cells). Duplication should be < 15 % for PCR-free WGS and < 25 % for WES; higher rates inflate apparent allele support and must be flagged.

4.3 Base-quality recalibration

Required for the GATK path (dbSNP + Mills + 1KG sites; confirm flattened residuals with AnalyzeCovariates). DeepTrio/DeepVariant are trained on un-recalibrated qualities and should be given the pre-BQSR BAM or run with BQSR disabled. Archive as reference-based CRAM after verifying round-trip checksums.

5. Stage 3 — Joint Trio Variant Calling

5.1 Per-sample GVCFs (corrected command)

```
# PCR-free library: disable the PCR indel model. Keep comments OFF continued lines.
gatk --java-options "-Xmx14g" HaplotypeCaller \
  -R ${REF} -I ${BAM} -O ${SM}.g.vcf.gz \
  -ERC GVCF \
  --pcr-indel-model NONE \
  -G StandardAnnotation -G StandardHCAAnnotation -G AS_StandardAnnotation \
  -A StrandBiasBySample \
  --max-alternate-alleles 6 \
  -L ${INTERVALS}
```

Do **not** add `--output-mode EMIT_VARIANTS_ONLY`: in GVCF mode it strips the reference blocks that prove each parent is hom-ref. `-A StrandBiasBySample` is optional: FORMAT/SB feeds the FS/SOR site annotations, but GenotypeGVCFs does not carry it into the joint VCF, so per-sample strand counts for Layer 2 come from a candidate-site pileup (§7.2).

WES/panels: restrict to targets ± 100 bp. WGS: scatter across ~50 interval shards. Sex chromosomes: calling male chrX/chrY non-PAR with `--sample-ploidy 1` is optional — `trio-dnm` treats diploid 1/1 calls there as hemizygous.

5.2 Joint genotyping

```
gatk GenomicsDBImport -V proband.g.vcf.gz -V father.g.vcf.gz -V mother.g.vcf.gz \
--genomicsdb-workspace-path trio_db -L ${INTERVALS} --merge-input-intervals
gatk GenotypeGVCFs -R ${REF} -V gendb://trio_db -O trio.vcf.gz \
-G StandardAnnotation -G AS_StandardAnnotation \
--only-output-calls-starting-in-intervals -L ${INTERVALS}
```

Joint genotyping is essential: it recovers low-confidence parental ALT evidence that distinguishes inherited variants from DNMs.

5.3 Site-level filtering and second caller

Apply VQSR/VETS when cohort-scale data exist; otherwise GATK hard filters:

Variant class	Filter expression
SNVs	QD < 2.0 QUAL < 30.0 FS > 60.0 MQ < 40.0 MQRankSum < -12.5 ReadPosRankSum < -8.0 SOR > 3.0
Indels	QD < 2.0 QUAL < 30.0 FS > 200.0 ReadPosRankSum < -20.0 SOR > 10.0

Failing sites keep their FILTER value (e.g. `QD2;QUAL30`) and fail Layer 2 of `trio-dnm call` (§7.2). Normalise (`bcftools norm -m -any -f ref.fa`) and annotate gnomAD v4.1 joint AF (`AF_joint`, `AF_grpmax_joint`), ClinVar and the in-house panel of normals (`PON_AF`) with `vcfanno` (`conf/vcfanno.toml`); `trio-dnm call` takes this annotated VCF. In parallel run DeepTrio + GLnexus.

Use the union of GATK and DeepTrio Mendelian-violation sites for candidate discovery. The union drives DeNovoGear and TrioDeNovo (§6), and the normalised, `vcfanno`-annotated DeepTrio VCF is passed to `trio-dnm call --extra-vcf`: its Mendelian-violation records that are absent from the GATK VCF go through the same cascade and are flagged `SECOND_ENGINE_ONLY`. They lack the GATK site annotations and `hiConfDeNovo`, so they need IGV review. Agreement between engines is counted as evidence in §6.5 rather than used as a pre-filter.

6. Stage 4 — Multi-Caller De Novo Detection

Each caller models the trio differently; agreement is the strongest single false-positive reducer. DeNovoGear and TrioDeNovo are run only at the candidate sites from §5.3, which keeps them tractable on WGS. The same sites are piled up from the three BAMs (`bcftools mpileup -a AD,ADF,ADR`) for the Layer-2 strand test (§7.2).

6.1 GATK CalculateGenotypePosteriors + PossibleDeNovo

```
gatk CalculateGenotypePosteriors -V trio.filtered.vcf.gz -ped trio.ped \
--skip-population-priors -O trio.cgp.vcf.gz
gatk VariantAnnotator -R ${REF} -V trio.cgp.vcf.gz -ped trio.ped \
-A PossibleDeNovo -O trio.cgp.dn.vcf.gz # writes hiConfDeNovo / loConfDeNovo
```

6.2 DeNovoGear

`dng dnm auto --ped trio.ped --vcf cand.vcf` on the joint-called candidate VCF (genotype likelihoods, PL), the same input as TrioDeNovo. Do not feed DeNovoGear a `bcftools mpileup` BCF: version 1.1.1 reads only the legacy samtools-0.1 BCF1 format. Keep `pp_dnm ≥ 0.5` as caller support (`bin/dng2tsv.py`).

6.3 TrioDeNovo

`triodenovo --ped trio.ped --in_vcf cand.vcf --out_vcf out.vcf --minDQ 7` on the joint-called candidate VCF (uses PL).

6.4 DeepTrio

Deep-learning trio caller (Kolesnikov 2021) with dedicated **Illumina WGS, Illumina WES and PacBio models** trained on GIAB trios; select `--model_type` by data type. Particularly strong for indels and low-complexity sequence. Mendelian-violation sites from the GLnexus-merged trio are the DeepTrio call set: they count as DeepTrio support (`--caller deeptrio=...`), and DeepTrio-only sites enter the cascade through `--extra-vcf` (§5.3). GLnexus writes BCF; convert it with `bcftools view -Oz`, because `trio-dnm` reads VCF only.

6.5 Consensus integration

Tiers are assigned only to candidates that pass every layer. Each carries `N_CALLERS` (the external callers reporting it) and the in-pipeline posterior (§7.5). `k` is the number of external callers supplied to `trio-dnm call` (`--caller`); the thresholds adapt to it:

Tier	<code>k ≥ 1</code> (external callers supplied)	<code>k = 0</code> (no external caller)	Expected use
HIGH	<code>N_CALLERS ≥ min(3, k)</code> and (<code>hiConfDeNovo</code> or posterior ≥ 0.95); mosaic and parental-mosaic tracks need posterior ≥ 0.95	Germline track only: posterior ≥ 0.95 and <code>hiConfDeNovo</code>	Primary set; IGV review before validation
MEDIUM	<code>N_CALLERS ≥ min(2, k)</code>	Posterior ≥ 0.95 or <code>hiConfDeNovo</code> ; mosaic and parental-mosaic tracks are capped here	Full IGV review before validation
LOW	Otherwise (<code>N_CALLERS</code> below the MEDIUM requirement)	Otherwise	Report only if biologically compelling and orthogonally validated

7. Stage 5 — Layered Filter Cascade

`trio-dnm call` takes the normalised, `vcfanno`-annotated joint VCF (§5.3), evaluates every Mendelian-violation candidate against **all** layers and records every failure reason in `candidates.tsv`; the waterfall in `call_summary.json` counts each candidate once, at the first layer it fails. Two read-level checks cannot be computed from a VCF and are covered elsewhere (§7.2). Defaults are for 30× PCR-free WGS; `--data-type wes|panel` applies the capture presets, and any threshold can be overridden with a JSON config (`--config`, containing only the keys you change).

7.1 Layer 1 — Genotype-level filters (ploidy-aware)

Metric	Proband	Each contributing parent	Rationale
Genotype	Carries ALT (0/1; hemizygous 1 or 1/1 where ploidy = 1)	Called hom-ref (0/0 or 0); a parent called 0/1 with significantly sub-heterozygous VAF goes to the parental-mosaic track (§9.4)	Mendelian violation
GQ	≥ 20	≥ 20	Genotype confidence
DP	≥ 10 and $\leq 2 \times$ sample mean	≥ 15 and $\leq 2 \times$ sample mean (≥ 30 in mosaic track)	$P(0 \text{ ALT reads} \mid \text{het, DP } 15) \approx 3 \times 10^{-5}$
DP, hemizygous father	—	chrX non-PAR (female proband) and chrY: ≥ 8 (<code>parent.min_dp_hemizygous</code> ; ≥ 15 in mosaic track; raised by the WES/panel presets)	His non-PAR X/Y depth is half the autosomal depth; a hemizygous carrier would show VAF ≈ 1
ALT reads	≥ 5	≤ 1 (0 in mosaic track)	Parental leakage
VAF	0.30–0.70 diploid; ≥ 0.85 hemizygous	≤ 0.05	Allele balance

Region (by build-specific PAR coordinates)	Proband ploidy	Parents that must be hom-ref
Autosomes, PAR1/PAR2	2	Father and mother
chrX non-PAR, female proband	2	Father (hemizygous ref) and mother
chrX non-PAR, male proband	1	Mother only (father transmits Y)
chrY non-PAR, male proband	1	Father only
chrM	—	Excluded (heteroplasmy pipeline)

Routing to the post-zygotic mosaic track requires **both** a proband VAF significantly below 0.5 (one-sided binomial $p < 10^{-3}$, $n = \text{ref} + \text{ALT reads}$) **and** $\text{VAF} < 0.40$ (`mosaic.max_vaf`). A low-VAF candidate whose test is not significant stays in the germline track, fails the heterozygous VAF range and is flagged `POSSIBLE_MOSAIC_NEEDS_DEPTH` for review. Hemizygous regions (male chrX/chrY non-PAR) use the same $\text{VAF} < 0.40$ limit with the binomial test against the hemizygous expectation; a heterozygous-looking VAF (0.40–0.70) there suggests sex-chromosome aneuploidy or a paralog artefact, so the candidate stays germline, fails the hemizygous VAF rule and is flagged `HEMIZYGOUS_HET_LIKE`. Candidates with significant low-level parental ALT reads, including parents genotyped 0/1 whose VAF is significantly sub-heterozygous, are routed to the parental-mosaic track (§9.4).

7.2 Layer 2 — Read and alignment quality

- Site FILTER must be PASS (or .): a FILTER set by the §5.3 hard filters or VQSR/VETS fails the site as `site_filter:<FILTER>` (config `site.honour_filter`).
- $\text{MQ} \geq 40$; $\text{FS} < 60$ (SNV) / < 200 (indel); $\text{SOR} < 3$ (SNV) / < 10 (indel).
- $\text{MQRankSum} \geq -12.5$; $\text{ReadPosRankSum} \geq -8$ (SNV) / ≥ -20 (indel).
- Proband ALT reads on both strands (≥ 1 each, ≥ 2 preferred) and per-sample strand Fisher $p \geq 10^{-3}$. GenotypeGVCFs does not carry FORMAT/SB into the joint VCF and HaplotypeCaller writes no ADF/ADR, so the strand counts come from `bcftools mpileup -a AD,ADF,ADR` of the three BAMs at the

candidate sites, passed as `trio-dnm call --strand-vcf`. FORMAT/SB or ADF/ADR in the joint VCF are used when present; with neither the candidate is flagged `NO_STRAND_INFO`.

- Soft-clipped read fraction < 30 % where an INFO key supplies it (`site.softclip_info_key`, default `SCF`).
- Not computed by `trio-dnm` from the VCF: indel support from ≥ 2 distinct read start positions (PCR stutter) and ALT reads concentrated in the terminal 10 % of read length. For germline candidates these are covered by ReadPosRankSum plus the mandatory IGV review before validation; for mosaic candidates by MosaicForecast's read-level features (§9.1).

7.3 Layer 3 — Genomic region exclusion (hard masks)

- ENCODE blacklist v2; GIAB stratifications: low mappability, segmental duplications, tandem repeats / low complexity; in-house recurrent-artefact and pseudogene masks.
- Homopolymer run ≥ 8 bp (SNV) or ≥ 6 bp (indel), computed from the reference around the site.
- Within ± 5 bp of a reference N stretch.

7.4 Layer 4 — Population and recurrence

- gnomAD v4.1 joint AF $< 1 \times 10^{-4}$, read from the `vcfanno` INFO key (`gnomAD_AF`; config `population.af_keys`). A ClinVar P/LP match is flagged `POP_AF_RESCUED_BY_CLINVAR` rather than removed: recurrent pathogenic DNMs at hypermutable CpGs, and CH variants in gnomAD blood donors, can appear at low counts.
- If no record of the input VCF carries a gnomAD AF key, this filter cannot act: `call` prints a warning and records `population_af_missing: true` in `call_summary.json`. Annotate with `vcfanno` first.
- In-house panel of normals (≥ 100 unrelated, identically processed): AF ≤ 0.01 , read from a `--pon` VCF or from the `vcfanno` INFO key `PON_AF` (`population.pon_info_key`).
- Recurrence in ≥ 5 unrelated trios in project history: removed as a likely artefact unless ClinVar P/LP (then flagged as a known recurrent pathogenic site).

7.5 Layer 5 — Bayesian trio posterior

`trio-dnm` computes the posterior from the joint-VCF genotype likelihoods (PL; a binomial read model with 1 % error when PL is absent), summing over every joint genotype configuration:

$$P(\text{DNM} \mid D) = \sum_{\{g_c > 0\}} P(D_f \mid \theta) P(D_m \mid \theta) P(D_c \mid g_c) \pi(\theta) \pi(\theta) T(g_c \mid \theta, \theta) / \sum_{\{g_f, g_m, g_c\}} (\text{same terms})$$

$\pi(g)$ Hardy–Weinberg prior with $q = \max(\text{gnomAD AF}, 1e-4)$ (floor keeps absent sites conservative)
 $T(g_c \mid g_f, g_m)$ Mendelian transmission with per-allele mutation $\mu = 1.2e-8$ (SNV) / $1e-9$ (indel)
 ploidy taken from the region table in §7.1 (hemizygous child: single transmitting parent)

- Retain candidates with posterior ≥ 0.80 ; the tiers use posterior ≥ 0.95 and `hiConfDeNovo` as set out in §6.5.
- For the mosaic tracks a two-class read model (true event at observed VAF vs sequencing error 0.5 %) replaces the diploid model, multiplied by the probability that each parent does not carry the event at that VAF.

7.6 Layer 6 — Call-set integrity (replaces v1.0 kinship-on-DNMs)

- **Parental leakage:** fraction of germline DNMs with any ALT read in each parent. An asymmetric excess (e.g. father 25 %, mother 3 %) points to contamination, a swap or under-sequencing of that parent.
- **Raw Mendelian error rate** (§3.2) recorded per trio with the number of informative sites; > 5 % over ≥ 1000 sites halts the run (stop-gate, §3.3).
- **Allele balance:** median proband VAF of germline DNMs should be 0.42–0.58 (`sanity.median_vaf_range`; checked by `annotate` when ≥ 10 DNMs).
- **Clustering:** DNMs within 20 kb are flagged `CLUSTERED_DNM` for review (complex events, MNVs, maternal-age-associated clusters, or mapping artefacts).

8. Stage 6 — Biological Sanity Checks

True germline DNMs reproduce known aggregate biology. `trio-dnm annotate` computes each metric and sets the trio status to PASS or REVIEW with explicit warnings. Deviation indicates a methodological problem until proven otherwise.

8.1 Expected counts

Data type	SNV DNMs	Indel DNMs	Hard alarm
WGS 30×	45–95 (callable-genome dependent)	3–12	< 30 or > 150
WES (coding), per trio	0–3	0–1	> 5
Targeted panel (~1 Mb), per trio	0–1	0–1	> 2

The WES and panel ranges are per-trio counts. The cohort mean is ≈ 1 –1.5 coding DNMs per exome, so about 30 % of normal exome trios have none: for WES and panels only counts above the range are flagged (REVIEW), and a count of zero is never an alarm.

Age adjustment (WGS): $\text{expected} \approx c + 1.51 \times \text{paternal age} + 0.37 \times \text{maternal age}$ (Jónsson 2017). The intercept c depends on the callable fraction; calibrate it on ≥ 20 in-house trios before interpreting the observed/expected ratio, which is reported per trio. A large excess with a distinctive spectrum should prompt review for germline hypermutation (e.g. paternal chemotherapy exposure; Kaplanis 2022).

8.2 Parent of origin

Phase with **unfazed** (read-backed and extended read-backed phasing using informative heterozygous sites). Expect 75–80 % paternal among phased SNV DNMs. Report the fraction with its Wilson 95 % CI and the number phased; alarm only when ≥ 10 are phased and the fraction lies outside **65–90 %**.

8.3 Mutation spectrum and signatures

- CpG transitions: 10–25 % of SNV DNMs (≈ 15 –20 % expected).
- Ti/Tv: 1.7–2.5 (WGS); 2.2–3.5 (WES).
- SBS-96 spectrum refit to SBS1 + SBS5 + SBS40a by non-negative least squares; report exposures and reconstruction cosine. COSMIC v3.4 split SBS40 into SBS40a/b/c; a requested name missing from the matrix is resolved with a warning (SBS40 $\rightarrow$ SBS40a). Unexpected signatures (e.g. SBS17, platinum SBS31/35) indicate artefact or exposure.

8.4 Indel properties

1–4 bp events should dominate; insertion:deletion $\approx 1:1$ in unique sequence. A deletion excess concentrated in repeats indicates homopolymer/STR leakage (tighten Layer 3).

9. Stage 7 — Mosaic-Aware Tracks

9.1 Coverage and callers

Reliable detection at VAF 0.05 needs $\geq 200\times$; standard $30\times$ WGS detects VAF ≥ 0.15 – 0.20 with modest sensitivity. For mosaic-focused studies deepen the proband by PCR-free targeted or duplex/UMI resequencing. Two complementary routes run in parallel:

- **Within `trio-dnm call`**: low-VAF candidates present in the joint VCF (routed as in §7.1), filtered with the mosaic thresholds below.
- **Mutect2 tumour-only** → **MosaicForecast** on the proband (SNVs with VAF 0.03–0.40, [Refine](#) model), then parental absence asserted from parental BAM pileups. MosaicForecast's read-level features (read position, base and mapping quality, clipping, strand) cover the read-position checks that `trio-dnm` cannot compute. DeepMosaic may be added for sensitivity at VAF 0.05–0.20.

9.2 Mosaic filters

- Proband ALT reads ≥ 3 and VAF ≥ 0.05 (0.02 for deep panels).
- Both parents: DP ≥ 30 (hemizygous father ≥ 15) and zero ALT reads.
- ALT on both strands (Fisher $p > 10^{-3}$; strand counts as in §7.2).
- ALT reads not concentrated in the terminal 10 % of read length: not computed by `trio-dnm`; covered by MosaicForecast on the Mutect2 route and by IGV review on the `trio-dnm call` route.

9.3 VAF classification

Use a one-sided binomial test of ALT reads against 0.5 and report the Wilson 95 % CI, instead of fixed VAF bins. Significantly sub-heterozygous VAF indicates post-zygotic origin; the lower the VAF, the later (or more lineage-restricted) the event. A non-significant result at 0.30–0.45 is compatible with a heterozygote and must not be called mosaic without deeper data; `trio-dnm call` therefore routes a candidate to the mosaic track only when the test is significant and VAF < 0.40 (§7.1).

9.4 Parental mosaicism (new)

A proband heterozygote whose parent shows ALT reads significantly above sequencing error (binomial $p < 10^{-4}$, ≥ 2 ALT reads, VAF ≤ 0.25) is **not** discarded; it is reported in the `parental_mosaic` track. This includes parents that the joint caller genotyped 0/1 but whose VAF is ≤ 0.25 and significantly below 0.5 (binomial $p < 10^{-3}$); such candidates are flagged `PARENT_CALLED_HET`. These variants carry substantial recurrence risk and should be confirmed by ddPCR or deep amplicon sequencing in parental blood, buccal and — for fathers — sperm.

9.5 Clonal haematopoiesis as a confounder

Blood-derived DNA from adult probands and from parents can carry CH clones (DNMT3A, TET2, ASXL1, PPM1D, TP53, JAK2, SF3B1, SRSF2 ...) at 2–35 % VAF. A low-VAF "DNM" in a CH gene in an adult

proband, or parental ALT reads at such a site, should be treated as CH until shown otherwise in a non-haematopoietic tissue. The CH screen (§11.7) is run on all trio members for this reason.

10. Stage 8 — Functional Annotation, Prediction and Prioritisation

10.1 VEP annotation (corrected command)

```
vep --input_file dnm.vcf --output_file dnm.vep.vcf --vcf --offline --cache \
--dir_cache ${VEP_CACHE} --dir_plugins ${VEP_PLUGINS} --assembly GRCh38 --fasta ${REF} \
--everything --mane --flag_pick_allele_gene \
--pick_order mane_select,mane_plus_clinical,canonical,appris,tsl,biotype,ccds,rank,length \
--plugin LoF,loftee_path:${LOFTEE},human_ancestor_fa:${ANC},conservation_file:${LOFTEE_SQL} \
--plugin CADD,snv=${CADD_SNV},indels=${CADD_INDEL} \
--plugin REVEL,file=${REVEL} \
--plugin AlphaMissense,file=${AM} \
--plugin SpliceAI,snv=${SAI_SNV},indel=${SAI_INDEL},cutoff=0.0 \
--plugin
dbNSFP,${DBNSFP},BayesDel_noAF_score,PrimateAI_score,ESM1b_score,EVE_score,MetaRNN_score,phyloP100way_
vertebrate,GERP++_RS \
--plugin UTRAnnotator,file=${UTR} --plugin NMD \
--custom
file=${GNOMAD},short_name=gnomADv4,format=vcf,type=exact,coords=0,fields=AF_joint%AF_grpmax_joint%nhom
alt_joint \
--custom
file=${CLINVAR},short_name=ClinVar,format=vcf,type=exact,coords=0,fields=CLNSIG%CLNREVSTAT%CLNDN \
--custom file=${CCRE_BED},short_name=ENCODE_cCRE,format=bed,type=overlap
```

`trio-dnm annotate` reports the MANE Select consequence (falling back to the VEP-flagged pick, then canonical) and keeps all other transcripts in the VCF for review. It reads the joint AF from the CSQ field `gnomADv4_AF_joint`. The SpliceAI plugin reads Illumina's pre-computed masked scores, which cover ± 50 bp only (§10.4). Pangolin is not part of this command; add `--plugin Pangolin` only if you install it. BP7 needs a phyloP score, and dbNSFP supplies one only for the variants it covers; add a genome-wide track (e.g. `--custom file=hg38.phyloP100way.bw,short_name=phyloP100way,format=bigwig`) to evaluate BP7 for synonymous variants.

10.2 Functional prediction toolkit

Question	Evidence / tool	Output column(s)
Does it truncate the protein and trigger NMD?	VEP consequence; LOFTEE HC/LC, flags and filters (END_TRUNC); NMD plugin (last exon / 50-nt rule)	consequence, loftee, loftee_flags, loftee_filter, nmd_escape
Is a missense change damaging?	AlphaMissense, REVEL, CADD v1.7, BayesDel, PrimateAI-3D, ESM1b, EVE, MetaRNN; concordance count	AlphaMissense, REVEL, ..., predictors_damaging
Does it alter splicing?	SpliceAI Δ (max of AG/AL/DG/DL; pre-computed ± 50 bp); Pangolin optional	spliceai
Does it disrupt a UTR element?	UTRAnnotator (uAUG gain, uORF stop loss, Kozak)	utr_annotator
Is the site conserved?	phyloP, GERP++	phyloP, GERP

Question	Evidence / tool	Output column(s)
Is it in a regulatory element?	ENCODE SCREEN cCREs (promoter/enhancer/CTCF)	cCRE
Where in the protein?	Pfam/InterPro domains; AlphaFold structure (manual) for 3-D context	domains, hgvsp
Is the gene intolerant?	gnomAD v4.1 pLI/LOEUF/missense Z; GeneBayes s_het; ClinGen HI	pLI, LOEUF, mis_z, shet, hi_score
Is it a known disease gene/variant?	OMIM/GenCC/G2P (inheritance, mechanism); ClinVar; hotspots (one- or three-letter protein notation)	disease, inheritance, mechanism, clinvar, hotspot

10.3 Calibrated in-silico evidence (PP3 / BP4)

Per ClinGen SVI, **one** pre-selected missense tool is used for PP3/BP4 to avoid double counting (default REVEL; configurable). Other predictors are reported for concordance and ranking only.

Tool	PP3 Supporting	PP3 Moderate	PP3 Strong	BP4 Supporting	BP4 Moderate	BP4 Strong	BP4 Very strong
REVEL (Pejaver 2022)	≥ 0.644	≥ 0.773	≥ 0.932	≤ 0.290	≤ 0.183	≤ 0.016	≤ 0.003
CADD PHRED (Pejaver 2022)	≥ 25.3	≥ 28.1	—	≤ 22.7	≤ 17.3	—	—
BayesDel noAF (Pejaver 2022)	≥ 0.13	≥ 0.27	≥ 0.41	≤ -0.18	≤ -0.36	—	—
AlphaMissense (Bergquist 2025)*	≥ 0.792	≥ 0.906	≥ 0.990	≤ 0.169	≤ 0.099	—	—
SpliceAI (Walker 2023)	≥ 0.20	—	—	≤ 0.10	—	—	—

Verify the AlphaMissense calibration values against the published table before clinical use; the calibration thresholds are configurable (`annotation.calibration` in `conf/defaults.json`). REVEL BP4 Very strong (≤ 0.003 , -8 points) follows the Pejaver 2022 intervals; a laboratory that caps computational evidence at Strong sets the fourth REVEL bp4 value to null. The AlphaMissense class cut-off (≥ 0.564 "likely pathogenic") is used for ranking only, not for ACMG strength.

10.4 Splicing

SpliceAI masked scores are evaluated for every variant class. The pre-computed Illumina files read by the VEP plugin were generated with a ± 50 bp window (GENCODE v24), so splice gains and losses 51–500 bp from the variant are not scored. For a ± 500 bp window run SpliceAI locally at the candidate sites (`spliceai -D 500`) or query the SpliceAI-lookup service. $\Delta \geq 0.20$ supports PP3 (not combined with a missense PP3); $\Delta \leq 0.10$ supports BP4 and, for synonymous variants at non-conserved sites (phyloP present and < 0.1), BP7. $\Delta \geq 0.50$ is treated as high confidence and prioritised for RT-PCR or minigene confirmation. Pangolin is an optional second opinion; it is not in the reference VEP command.

10.5 Non-coding DNMs

Non-coding DNMs ($\approx 98\%$ of WGS DNMs) are reported as Tier 5 and ranked by cCRE overlap, conservation (phyloP), CADD and proximity to phenotype-relevant genes. Interpretation is research-grade unless supported by functional data (MPRA, reporter assays, CRISPRi).

10.6 Gene constraint

- LoF-intolerant: $pLI \geq 0.9$, gnomAD v4 LOEUF < 0.6 , $s_{het} \geq 0.1$, or ClinGen HI score 3.
- Missense-constrained: missense $Z \geq 3.09$. Regional missense constraint and constrained coding regions (CCRs) are recommended manual follow-ups for high-ranked missense DNMs.

10.7 Provisional ACMG/AMP classification

Only criteria derivable from annotation are automated; each is scored with the Bayesian point system (Tavtigian 2020: Supporting 1, Moderate 2, Strong 4, Very strong 8; benign criteria negative). Class: ≥ 10 Pathogenic; 6–9 Likely pathogenic; 0–5 VUS; -1 to -6 Likely benign; ≤ -7 Benign; BA1 stand-alone Benign.

Criterion	Automated rule
PVS1	Null variant (stop, frameshift, canonical splice, start loss) in a gene with LoF disease mechanism (HI 3, constraint as §10.6, or curated mechanism). Very strong; Strong if predicted NMD escape (NMD plugin or LOFTEE <code>END_TRUNC</code>); Moderate for start loss; LOFTEE LC downgrades exactly one level (Very strong $\rightarrow$ Strong, Strong $\rightarrow$ Moderate, Moderate $\rightarrow$ Supporting). Full Abou Tayoun decision tree is manual.
PS2 / PM6	Germline DNM: PS2 when parentage is confirmed (Stage 1) and the variant is orthogonally validated; otherwise PM6. Phenotype-specificity weighting (ClinGen SVI de novo points) is manual.
PM2_Supporting	Absent or $AF \leq 1 \times 10^{-5}$ in gnomAD v4.1.
PP2	Missense in a missense-constrained gene ($Z \geq 3.09$).
PP3 / BP4	Calibrated single tool (§10.3; REVEL BP4 up to Very strong) or SpliceAI; never both for the same variant.
BS1 / BA1	$AF > 1 \times 10^{-3}$ / > 0.05 .
BP7	Synonymous, SpliceAI ≤ 0.10 , phyloP present and < 0.1 (no BP7 without a phyloP score).

Automated classes are provisional and are labelled as such in every output. Segregation, functional evidence (PS3/BS3), phenotype specificity and case-level evidence require a qualified reviewer.

10.8 Prioritisation tiers

Tier	Criteria
Tier 1	Provisional P/LP; or ClinVar P/LP; or curated hotspot (protein changes compared after normalising one- and three-letter notation, so p.R132H matches p.Arg132His); or null variant in a LoF-intolerant or known dominant (AD/XLD) disease gene
Tier 2	Missense/in-frame with PP3 $\geq$ Moderate (or AlphaMissense ≥ 0.564) in a missense-constrained, LoF-intolerant or known dominant disease gene
Tier 3	Predicted splice-altering (SpliceAI ≥ 0.20) outside the canonical donor/acceptor sites; canonical splice donor/acceptor variants go to Tier 4 unless they reach Tier 1

Tier	Criteria
Tier 4	Other coding and splice-region variants, including canonical splice-site variants that miss Tier 1
Tier 5	Non-coding; reported with cCRE / conservation context

10.9 Phenotype integration and ranking

Run Exomiser or LIRICAL with the proband HPO terms and supply per-gene phenotype scores (0–1). `trio-dnm` ranks variants by a transparent heuristic score = ACMG points + DNM-tier bonus (HIGH 2, MEDIUM 1) + constraint bonus (1) + 4 × phenotype score + tier bonus (Tier 1: 4 ... Tier 4: 1). The score orders review; it is not a probability. Typical output: 1–5 shortlisted candidates per clinical trio.

10.10 Cohort-level analyses

- Gene-level DNM enrichment against mutation-rate models: DeNovoWEST (Kaplanis 2020) or denovolyzeR (Samocha 2014 framework); report genome-wide significance thresholds for the number of genes tested.
- Gene-set / pathway enrichment of genes with damaging DNMs versus synonymous DNMs as an internal control.
- Cohort SBS-96 spectra and signature refit; 3-D clustering of missense DNMs on AlphaFold structures for recurrently hit genes.

11. Stage 9 — Somatic Mutation Analysis (new)

Somatic analysis answers three distinct questions. All three use the trio as a "super-normal": an ALT allele present in either parent is inherited, not somatic, even where the matched normal is shallow.

Use case	Samples	Question
A. Tumour–normal	Proband tumour + proband blood (+ parents)	Somatic drivers, mutational processes, and whether a germline DNM or inherited variant is the first hit
B. Affected tissue vs blood	Lesional tissue (brain, skin, vascular, overgrowth) + blood	Tissue-restricted post-zygotic variants (e.g. PIK3CA, MTOR, GNAQ, KRAS) that blood-based trio analysis cannot see
C. Clonal haematopoiesis	Blood of every trio member	CH clones that masquerade as mosaic DNMs or parental leakage (§9.5)

11.1 Calling (use cases A and B)

Run three independent callers on the tumour/tissue–blood pair and integrate by support (default: ≥ 2 callers):

```

gatk Mutect2 -R ${REF} -I tumor.bam -I normal.bam -normal ${NORMAL} \
  --germline-resource af-only-gnomad.hg38.vcf.gz --panel-of-normals pon.vcf.gz \
  --f1r2-tar-gz f1r2.tar.gz -O unfiltered.vcf.gz
gatk LearnReadOrientationModel -I f1r2.tar.gz -O rom.tar.gz
gatk GetPileupSummaries -R ${REF} -I tumor.bam -V common.vcf.gz -L common.vcf.gz -O t.table
gatk GetPileupSummaries -R ${REF} -I normal.bam -V common.vcf.gz -L common.vcf.gz -O n.table
gatk CalculateContamination -I t.table -matched n.table -O contam.table --tumor-segmentation seg.table
gatk FilterMutectCalls -R ${REF} -V unfiltered.vcf.gz --contamination-table contam.table \
  --tumor-segmentation seg.table --ob-priors rom.tar.gz -O filtered.vcf.gz
# Manta (own step) -> candidateSmallIndels.vcf.gz -> Strelka2 --indelCandidates
# DeepSomatic (WGS/WES model) is the third caller

```

11.2 Somatic filters (trio-dnm somatic)

Metric	Threshold	Rationale
Tumour/tissue depth, ALT reads, VAF	DP $\geq$ 20, ALT $\geq$ 4, VAF $\geq$ 0.05	Minimum evidence
Matched normal	DP $\geq$ 10, ALT $\leq$ 1, VAF $\leq$ 0.02	Excludes germline and normal contamination
Parents (trio super-normal)	< 2 ALT reads in each parent	Removes inherited variants missed in a shallow normal
Population AF	gnomAD $\leq 1 \times 10^{-3}$ (INFO key, else the gnomAD AF in the VEP CSQ)	Removes germline polymorphisms
Caller support	≥ 2 of Mutect2 / Strelka2 / DeepSomatic	Consensus
FFPE (with --ffpe)	Only C>T / G>A SNVs at VAF $\leq$ 0.10 are flagged (ffpe_suspect), not filtered	Deamination artefact; Mutect2's orientation-bias model filters separately

`somatic` and `ch-screen` validate sample IDs: a tumour, normal, parent or CH sample missing from the VCF header stops the run with an error instead of producing empty results.

11.3 Copy number, purity and clonality

- Allele-specific copy number with CNVkit (`call -m clonal` with SNP B-allele frequencies) or PURPLE/FACETS; purity/ploidy from the same tool or pathology.
- Cancer-cell fraction: $CCF = VAF \times (p \cdot CN_t + 2(1 - p)) / (p \cdot m)$, with purity p and local total copy number CN_t . The multiplicity is estimated per variant as $m = \text{clamp}(\text{round}(VAF \cdot (p \cdot CN_t + 2(1 - p)) / p), 1, \text{major CN})$. CCF is capped at 1 (`ccf`); the uncapped value is reported as `ccf_raw`. A variant is **clonal** when the upper 95 % bound of CCF ≥ 0.8 .
- LOH (minor CN = 0; copy-neutral or deletion) is recorded for each somatic variant (`loh` column) and at each germline hit position for two-hit analysis (§11.5).

11.4 Mutational burden, signatures and MSI

- TMB = PASS non-synonymous somatic SNVs/indels per callable coding Mb (default 30 Mb for exomes; set to your callable territory). TMB-high $\geq 10/\text{Mb}$.
- SBS-96 spectrum and NNLS refit to COSMIC v3.4, with reconstruction cosine. Below 50 PASS SNVs a full-catalogue fit over-fits, so the refit is skipped unless a biologically plausible subset is given (`--signature-subset`).

- MSI with MSIsensor-pro on the pair; mismatch-repair or POLE signatures (SBS6/15/26/44, SBS10a/b) cross-checked against germline MMR/POLE findings.

11.5 Driver and two-hit analysis

- Drivers: curated hotspots (cancerhotspots, COSMIC) and damaging variants in COSMIC Cancer Gene Census genes; OncoKB/CIViC levels where licensed.
- Two-hit detection:** a gene with a Tier 1/2 or P/LP germline variant (from §10: de novo, or inherited when an annotated inherited-variant table is supplied) plus a PASS somatic SNV/indel in the same gene and/or LOH (minor CN = 0) at the germline variant position is reported as a candidate Knudson second hit (`<umor>.two_hit.tsv`) — the key link between the trio germline result and the tumour. Germline + LOH is reported even without a somatic point mutation; LOH alone, without a germline hit, is recorded (`loh` column) but is not a second hit.

11.6 Affected tissue vs blood (use case B)

The same paired workflow is run with `--mode tissue`; PASS variants are labelled `SOMATIC_MOSAIC`. Expect low VAF (often 1–10 %): deepen to $\geq 500\times$ with targeted/UMI sequencing of candidate genes, and confirm by ddPCR in a second piece of lesional tissue.

11.7 Clonal haematopoiesis screen (use case C)

Mutect2 in tumour-only mode on each blood sample, restricted to a CH gene panel, with PoN and germline resource, then VEP; `trio-dnm ch-screen` reports HIGH/MODERATE-impact variants (from the VEP CSQ, preferred) in CH genes at VAF 2–35 % (≥ 3 ALT reads) with Wilson CIs. Without a CSQ header the gene is read from an INFO key (`--gene-info-key`, default `GENE`), impact cannot be assessed and the impact filter is skipped, with a warning in the summary. Hits are cross-referenced with the DNM and mosaic tracks.

11.8 Somatic validation

Validate reportable somatic variants by an orthogonal method (amplicon-UMI sequencing or ddPCR), in a second tumour region where available, and in the matched normal to confirm absence.

12. Stage 10 — Orthogonal Validation

Variant class	Primary modality	Notes
Germline SNV (VAF ≈ 0.5)	Sanger of trio amplicons	Sensitivity floor ~15–20 % VAF
Indel ≤ 20 bp	Sanger (proband + parents)	Confirm length and breakpoints
Post-zygotic SNV (VAF 0.02–0.30)	ddPCR or amplicon-UMI sequencing	Sanger insufficient
Parental mosaic	ddPCR / deep amplicon in blood, buccal and sperm	Recurrence-risk counselling
Splice-altering	RT-PCR + Sanger of cDNA, or minigene	Choose a tissue that expresses the gene
Recurrent / clustered / low-mappability	PacBio HiFi or ONT long reads	Rules out reference / paralog artefacts; phases complex events

Variant class	Primary modality	Notes
Somatic	Amplicon-UMI or ddPCR in tumour and normal	Independent tumour region where possible

Suspected post-zygotic events are tested in at least two tissues of distinct embryonic origin (e.g. blood and buccal/saliva, or affected vs unaffected tissue). For autosomal-dominant disease genes, deep paternal sperm sequencing ($\geq 1000\times$) quantifies gonadal mosaicism and recurrence risk.

13. Reporting and Documentation

13.1 Outputs

Output	File (reference implementation)
QC stop-gate verdict	<fam>.qc_gate.json, somalier HTML, MultiQC report
All Mendelian-violation candidates with every failure reason and flag	<fam>.candidates.tsv
Filter waterfall, tiers, leakage, raw Mendelian error rate and gate, <code>population_af_missing</code> , effective configuration	<fam>.call_summary.json
Passing DNMs (all tracks, tiered, INFO-annotated)	<fam>.dnm.vcf
Fully annotated, ranked variant table	<fam>.annotated.tsv
Clinical shortlist (Tier 1–3)	<fam>.shortlist.tsv
Sanity checks, signatures, class/tier counts	<fam>.qc.json, <fam>.sbs96.tsv
Interactive report (waterfall, spectrum, ranked table)	<fam>.report.html
Mosaic candidates with parental pileups	<fam>.mosaic_candidates.tsv
Somatic calls, CCF, drivers, two-hits, TMB, signatures	<tumor>.somatic.tsv, <tumor>.two_hit.tsv, <tumor>.somatic_summary.json
CH screen	<sample>.ch_screen.tsv, <sample>.ch_summary.json
Provenance	Nextflow trace/timeline/report/DAG and a provenance JSON in <code>results/pipeline_info/</code> ; effective <code>trio-dnm</code> configuration in each <code>call_summary.json</code>

13.2 Quality-assurance log (per trio)

- Stage-1 outcomes (FREEMIX, relatedness/IBS0, sex, depth, DNA source), the Mendelian-error gate, and any overrides (`--no-halt`) with justification.
- Waterfall counts per layer; sanity-check status and every warning.
- Validation modality and outcome for each reported variant; reviewer initials and date for each Tier 1–2 candidate and each ACMG class change.

13.3 Data deposition

Raw data to controlled-access archives (dbGaP, EGA) under appropriate consent. Clinically relevant variants to ClinVar; research DNMs to community resources (denovo-db, GeneMatcher/Matchmaker Exchange) after de-identification.

14. Troubleshooting

Symptom	Likely cause	Action
<code>trio-dnm call</code> exits with code 3	Raw Mendelian error rate > 5 %: swap, contamination or wrong PED	Check somalier relatedness, sex and FREEMIX; use <code>--no-halt</code> only after documented review
<code>population_af_missing</code> in <code>call_summary.json</code>	<code>call</code> was given a VCF without gnomAD INFO	Run <code>vcfanno</code> (<code>conf/vcfanno.toml</code>) and call on the annotated VCF
Many candidates flagged <code>NO_STRAND_INFO</code>	Joint VCF has no SB/ADF/ADR and no <code>--strand-vcf</code> given	Pass the candidate-site <code>bcftools mpileup -a AD,ADF,ADR</code> VCF with <code>--strand-vcf</code>
WGS DNM count > 150	Permissive filters; contamination; LCL DNA	Check FREEMIX and parental leakage; raise parental DP to 20 and ALT to 0; confirm Layer 3 masks applied
WGS DNM count < 30	Over-filtering; low parental depth; GVCF reference blocks missing	Inspect the waterfall; confirm HaplotypeCaller did not use <code>EMIT_VARIANTS_ONLY</code> ; relax proband VAF floor to 0.25
Paternal fraction < 65 % (≥ 10 phased)	Maternal-side artefact, contamination or swap	Re-run somalier; IGV-review 20 random DNMs
One parent shows ALT reads at many DNMs	Contamination of the <i>other</i> sample or under-sequenced parent	Compare per-parent leakage fractions and FREEMIX
Many low-VAF DNMs in DNMT3A/TET2/ASXL1	Clonal haematopoiesis	Run the CH screen; confirm in buccal/saliva DNA
Recurrent variants across trios	Mappability, paralog or reference artefact	Add to the in-house recurrence file and mask
Excess indel DNMs in repeats	Homopolymer/STR leakage	Verify homopolymer and tandem-repeat masks; tighten indel filters
chrX DNMs missing in male probands	Hemizygous calls treated as diploid	Check pedigree sex; <code>trio-dnm</code> applies ploidy by PAR coordinates of the chosen build
DeNovoGear parse errors or no output	Output format differs between builds; BCF input from modern bcftools	Run on the candidate VCF (<code>--vcf</code> , §6.2); check <code>bin/dng2tsv.py</code> aliases; fall back to 3-caller consensus
Mosaic candidates fail Sanger	VAF below Sanger floor	Confirm by ddPCR or amplicon-UMI sequencing
Many FFPE-flagged somatic calls	Formalin deamination	Rely on orientation-bias model; require ≥ 2 callers; validate
Out-of-memory in joint genotyping	Too few shards	Scatter over more intervals; raise memory label

15. Reference Implementation Map

The repository `trio_dnm_pipeline/` implements this protocol. The Python engine is dependency-free and covered by an automated test suite on a synthetic trio (true DNM, parental leakage, parental mosaic, proband mosaic, common variant, masked region, homopolymer, low MQ, inherited, male chrX hemizygous DNM), annotation/ACMG scenarios, somatic, CH, QC-gate and parser tests. Every Nextflow process has a stub block; stub dry runs on bundled placeholder inputs (`nextflow run main.nf -stub -profile test_stub`, and with `-params-file tests/nextflow/params_fastq.yaml` for FASTQ input) complete without failures on Nextflow 24.10.4. The workflow has not yet been executed with the real tools and must be validated on GIAB trio HG002/HG003/HG004 before production use. `trio-dnm` tasks stage the `trio_dnm` package into each task directory and put it on `PYTHONPATH`, so they run in any Python ≥ 3.10 image (default `python:3.11`) or in the image built from the provided Dockerfile. Every resource file (with any index next to it) is staged into the tasks as a Nextflow `path` input, so it is visible inside Docker/Singularity containers without bind mounts (see the header of `nextflow.config`).

Stage (§)	Nextflow process(es)	trio-dnm command
1 QC (§3)	MOSDEPTH, VERIFYBAMID2, SOMALIER_EXTRACT/RELATE, QC_GATE; MultiQC report at the end of the run	<code>qc-gate</code>
2 Read processing (§4; FASTQ input only)	ALIGN subworkflow (<code>modules/align.nf</code>): fastp, FastQC, BWA-MEM2 with full @RG, MarkDuplicates, BQSR	—
3 Joint calling (§5)	HAPLOTYPECALLER, GENOTYPE_TRIO (hard filter, CGP, PossibleDeNovo), NORMALISE, VCFANNO, DEEPTRIO, GLNEXUS	—
4–5 DNM detection + cascade (§6–7)	CANDIDATE_SITES, MPILEUP_CANDIDATES (strand pileup), DENOVOGEAR, TRIODENOVO, TRIO_DNM_CALL	<code>call</code>
7 Mosaic (§9)	MUTECT2_PROBAND, MOSAICFORECAST, PARENTAL_ABSENCE	<code>call</code> (mosaic tracks)
6, 8 Annotation + sanity (§8, §10)	VEP, UNFAZED, TRIO_DNM_ANNOTATE	<code>annotate</code> , <code>signatures</code>
9 Somatic (§11)	MUTECT2_PAired, MANTA, STRELKA2, DEEPSOMATIC, CNVKIT, VEP_SOMATIC, TRIO_DNM_SOMATIC	<code>somatic</code>
9 CH screen (§11.7)	MUTECT2_CH, VEP_CH, TRIO_DNM_CH	<code>ch-screen</code>
10 Orthogonal validation (§12)	Laboratory work; not automated	—
Provenance (§13.1)	Nextflow reports and provenance JSON in <code>results/pipeline_info/</code>	<code>config</code>

```
# post-calling only. trio.anno.vcf.gz = normalised joint VCF annotated by vcfanno
# (gnomAD_AF, PON_AF, ClinVar); strand pileup of the candidate sites for Layer 2:
bcftools mpileup -f GRCh38.fa -R cand_sites.tsv -a AD,ADF,ADR,DP -Oz \
    -o FAM001.strand.vcf.gz P.bam F.bam M.bam
trio-dnm call --vcf trio.anno.vcf.gz --ped trio.ped --fasta GRCh38.fa \
    --exclude-bed masks.bed --strand-vcf FAM001.strand.vcf.gz \
    --extra-vcf deeptrio.anno.vcf.gz --caller deeptrio=dt.dnm.vcf.gz \
    --caller triodenovo=tdn.vcf --out FAM001
trio-dnm annotate --vcf FAM001.vep.vcf --call-summary FAM001.call_summary.json \
    --gene-table genes.tsv --fasta GRCh38.fa \
    --signatures COSMIC_v3.4_SBS_GRCh38.txt \
    --paternal-age 34 --maternal-age 31 \
    --parent-of-origin FAM001.poo.bed --out FAM001
trio-dnm somatic --vcf T.mutect2.vep.vcf --tumor T --normal P \
    --trio-vcf trio.vcf.gz --father F --mother M \
    --caller strelka2=snvs.vcf.gz,indels.vcf.gz \
    --caller deepsomatic=ds.vcf.gz \
    --segments T.segments.tsv --purity 0.6 \
    --germline-annotated FAM001.annotated.tsv --out T
# workflow: stub dry run on bundled placeholder inputs, then a real run
nextflow run main.nf -stub -profile test_stub
nextflow run main.nf -profile docker -params-file params.yaml
```

16. Key References

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